

PSEUDO CODE

START

Display Login Page

IF user enters valid login details THEN

 Identify user role

 IF role = Student THEN

 Open Student Dashboard

 Dashboard displays:

- Learning modules
- Quiz score
- Progress
- Points
- Badges
- Upcoming mock scenario

 IF user selects Learning Modules THEN

 Display disaster modules

 IF user selects a module THEN

 Display lesson

 Display safety information

 Display progress

 END IF

END IF

IF user selects Quiz THEN

 Display quiz questions

 FOR each question

 Display multiple-choice answers

 Accept student's answer

 END FOR

 Calculate score

 Store/display score

 Display quiz result

 IF score $\geq$ passing score THEN

 Award badge

 END IF

END IF

IF user selects Mock Scenario THEN

 Display emergency situation

 Display possible actions

 Accept student's choice

 Display correct/incorrect feedback

 Display safety explanation

END IF

IF user selects Progress THEN

 Display:

- Module progress
- Quiz performance
- Completion percentage

END IF

IF user selects Badges THEN

Display earned and locked badges

END IF

ELSE IF role = Teacher/Admin THEN

Open Admin Dashboard

Display:

- Number of students
- Average quiz score
- Module completion
- Student activity
- Reports

END IF

ELSE

Display "Invalid username or password"

END IF

IF user clicks Logout THEN

End session

Return to Login Page

END IF

END

SOURCE CODE

```
from flask import Flask, render_template, request, redirect, url_for, session
```

```
app = Flask(__name__)
```

```
app.secret_key = "disastered-secret-key"
```

```
# -----
```

```
# DEMO USERS
```

```
# -----
```

```
users = {  
    "student@disastered.com": {  
        "password": "1234",  
        "name": "John Doe",  
        "role": "Student"  
    },
```

```
    "admin@disastered.com": {  
        "password": "admin",  
        "name": "Admin User",  
        "role": "Admin"  
    }  
}
```

```
# -----
```

```
# DISASTER MODULES
```

```
modules = [  
  {  
    "id": 1,  
    "name": "Earthquakes",  
    "icon": "bi-house-exclamation",  
    "description": "Learn what to do before, during and after an earthquake.",  
    "progress": 75,  
    "difficulty": "Easy"  
  },  
  
  {  
    "id": 2,  
    "name": "Floods",  
    "icon": "bi-water",  
    "description": "Understand flood risks and important safety procedures.",  
    "progress": 40,  
    "difficulty": "Easy"  
  },  
  
  {  
    "id": 3,  
    "name": "Fire Safety",  
    "icon": "bi-fire",  
    "description": "Learn how to prevent and respond to fire emergencies.",  
    "progress": 60,  
    "difficulty": "Medium"
```

```
},
```

```
{
```

```
  "id": 4,
```

```
  "name": "First Aid",
```

```
  "icon": "bi-heart-pulse",
```

```
  "description": "Learn basic first aid skills for emergency situations.",
```

```
  "progress": 20,
```

```
  "difficulty": "Medium"
```

```
},
```

```
{
```

```
  "id": 5,
```

```
  "name": "Cyclones",
```

```
  "icon": "bi-wind",
```

```
  "description": "Learn how to prepare for strong winds and cyclones.",
```

```
  "progress": 30,
```

```
  "difficulty": "Medium"
```

```
},
```

```
{
```

```
  "id": 6,
```

```
  "name": "Tsunamis",
```

```
  "icon": "bi-tsunami",
```

```
  "description": "Understand tsunami warnings and evacuation procedures.",
```

```
  "progress": 10,
```

```
  "difficulty": "Hard"
```

```
}
```

```
]
```

```
# -----
```

```
# QUIZ QUESTIONS
```

```
# -----
```

```
quiz_questions = [
```

```
{
```

```
    "question": "What should you do during an earthquake?",
```

```
    "options": [
```

```
        "Run towards the window",
```

```
        "Drop, Cover and Hold On",
```

```
        "Use the elevator",
```

```
        "Stand outside immediately"
```

```
    ],
```

```
    "answer": 1
```

```
},
```

```
{
```

```
    "question": "Which place is safer during an earthquake?",
```

```
    "options": [
```

```
        "Under a sturdy table",
```

```
        "Near a window",
```

```
        "On a balcony",
```

```
        "Inside an elevator"
```

```
    ],
```

```
    "answer": 0
```

```
},
```

```
{
```

```
  "question": "What should you do after an earthquake?",
```

```
  "options": [
```

```
    "Ignore warnings",
```

```
    "Check for injuries and hazards",
```

```
    "Return immediately to damaged buildings",
```

```
    "Use damaged electrical equipment"
```

```
  ],
```

```
  "answer": 1
```

```
},
```

```
{
```

```
  "question": "Which item is useful in an emergency kit?",
```

```
  "options": [
```

```
    "Flashlight",
```

```
    "Gaming console",
```

```
    "Television",
```

```
    "Decorative items"
```

```
  ],
```

```
  "answer": 0
```

```
},
```

```
{
```

```
  "question": "What does 'Drop, Cover and Hold On' mean?",
```

```
  "options": [
```

```
    "Drop to the ground, take cover and hold on",
```



```
        "Run outside",
        "Call friends",
        "Open all windows"
    ],
    "answer": 0
}
]
```

```
# -----
# LOGIN
# -----
```

```
@app.route("/", methods=["GET", "POST"])
def login():
```

```
    if request.method == "POST":
```

```
        email = request.form.get("email")
        password = request.form.get("password")
```

```
        user = users.get(email)
```

```
        if user and user["password"] == password:
```

```
            session["email"] = email
            session["name"] = user["name"]
            session["role"] = user["role"]
```

```
if user["role"] == "Admin":  
    return redirect(url_for("admin_dashboard"))
```

```
return redirect(url_for("dashboard"))
```

```
return render_template(  
    "login.html",  
    error="Invalid email or password"  
)
```

```
return render_template("login.html")
```

```
# -----
```

```
# STUDENT DASHBOARD
```

```
# -----
```

```
@app.route("/dashboard")
```

```
def dashboard():
```

```
    if "email" not in session:  
        return redirect(url_for("login"))
```

```
    return render_template(  
        "dashboard.html",  
        modules=modules,  
        name=session["name"]
```

```
)
```

```
# -----
```

```
# MODULES
```

```
# -----
```

```
@app.route("/modules")
```

```
def module_list():
```

```
    if "email" not in session:
```

```
        return redirect(url_for("login"))
```

```
    return render_template(
```

```
        "modules.html",
```

```
        modules=modules
```

```
)
```

```
# -----
```

```
# MODULE DETAIL
```

```
# -----
```

```
@app.route("/module/<int:module_id>")
```

```
def module_detail(module_id):
```

```
    module = next(
```

```
        (m for m in modules if m["id"] == module_id),
```

None

)

if not module:

return redirect(url_for("module_list"))

return render_template(

"module_detail.html",

module=module

)

QUIZ

@app.route("/quiz", methods=["GET", "POST"])

def quiz():

if request.method == "POST":

score = 0

for i, question in enumerate(quiz_questions):

answer = request.form.get(f"question_{i}")

if answer is not None:

```
if int(answer) == question["answer"]:
```

```
    score += 1
```

```
percentage = int(
```

```
    (score / len(quiz_questions)) * 100
```

```
)
```

```
session["quiz_score"] = percentage
```

```
return redirect(url_for("quiz_result"))
```

```
return render_template(
```

```
    "quiz.html",
```

```
    questions=quiz_questions
```

```
)
```

```
# -----
```

```
# QUIZ RESULT
```

```
# -----
```

```
@app.route("/quiz-result")
```

```
def quiz_result():
```

```
    score = session.get("quiz_score", 0)
```

```
    badge = score >= 80
```

```
return render_template(
    "quiz_result.html",
    score=score,
    badge=badge
)
```

```
# -----
# MOCK SCENARIOS
# -----
```

```
@app.route("/scenarios")
def scenarios():

    return render_template("scenarios.html")
```

```
# -----
# PROGRESS
# -----
```

```
@app.route("/progress")
def progress():

    return render_template(
        "progress.html",
        modules=modules
    )
```

```
# -----
```

```
# BADGES
```

```
# -----
```

```
@app.route("/badges")
```

```
def badges():
```

```
    badges_list = [
```

```
        {
```

```
            "name": "Earthquake Ready",
```

```
            "icon": "bi-shield-fill-check",
```

```
            "earned": True
```

```
        },
```

```
        {
```

```
            "name": "Fire Safety Hero",
```

```
            "icon": "bi-fire",
```

```
            "earned": True
```

```
        },
```

```
        {
```

```
            "name": "Flood Awareness",
```

```
            "icon": "bi-water",
```

```
            "earned": True
```

```
        },
```

```
        {
```

```
            "name": "First Aid Beginner",
```

```
            "icon": "bi-heart-pulse",
```

```
        "earned": False
    },
    {
        "name": "Quiz Master",
        "icon": "bi-trophy-fill",
        "earned": False
    }
]
```

```
return render_template(
    "badges.html",
    badges=badges_list
)
```

```
# -----
# CERTIFICATES
# -----
```

```
@app.route("/certificates")
def certificates():
```

```
    return render_template(
        "certificates.html"
    )
```

```
# -----
```



```
# RESOURCES
```

```
# -----
```

```
@app.route("/resources")
```

```
def resources():
```

```
    return render_template(
```

```
        "resources.html"
```

```
    )
```

```
# -----
```

```
# ADMIN DASHBOARD
```

```
# -----
```

```
@app.route("/admin")
```

```
def admin_dashboard():
```

```
    if session.get("role") != "Admin":
```

```
        return redirect(url_for("login"))
```

```
    return render_template(
```

```
        "admin_dashboard.html"
```

```
    )
```

```
# -----
```

```
# LOGOUT
```

```
# -----
```

```
@app.route("/logout")
```

```
def logout():
```

```
    session.clear()
```

```
    return redirect(url_for("login"))
```

```
# -----
```

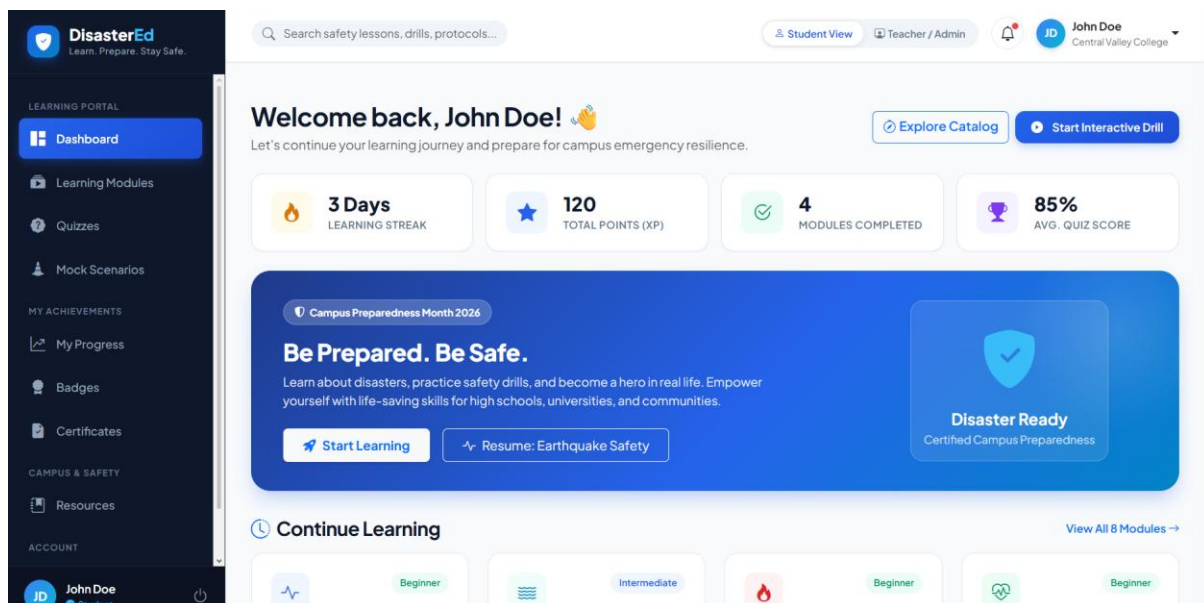
```
# RUN APPLICATION
```

```
# -----
```

```
if __name__ == "__main__":
```

```
    app.run(debug=True)
```

SCREENSHOTS OF OUTPUT



LEARNING MODULE

DisasterEd

Learn. Prepare. Stay Safe.

LEARNING PORTAL

Dashboard

Learning Modules

Quizzes

Mock Scenarios

MY ACHIEVEMENTS

My Progress

Badges

Certificates

CAMPUS & SAFETY

Resources

ACCOUNT

JohnDoe

Student

Search safety lessons, drills, protocols...

Student ViewTeacher / Admin

JohnDoeCentral Valley College

Disaster Learning Modules

Interactive, scientifically verified curricula designed for primary, secondary, and higher education campus safety.

8 Modules Available

All CategoriesGeological HazardsWeather & StormsFire & StructuralFirst Aid & Health

Curriculum Standard: FEMA / Red Cross Aligned

Beginner

Earthquake

Master seismic tremors, structural vulnerability assessment, and the vital "Drop, Cover, and Hold On" drill.

20 Mins75% Completed

Continue Learning

Intermediate

Flood

Understand flood warning signals, dangerous flash surge currents, safe high ground, and contamination hazards.

25 Mins40% Completed

Continue Learning

Beginner

Fire

Learn fire hazard mitigation, the PASS fire extinguisher technique, crawling under smoke, and assembly routines.

15 Mins60% Completed

Continue Learning

Intermediate

Cyclone

Analyze meteorological track maps, severe barometric pressure drops, and reinforce interior shelter spaces.

20 Mins0% Completed

Start Learning

Advanced

Advanced

Beginner

Beginner

Dashboard / Modules / Earthquake Safety

Geological Safety

Beginner Level

Estimated: 20 Mins

Earthquake Safety & Structural Preparedness

Lesson Progress 75%

Learn how seismic fault movements occur, how to mitigate hazards before ground shaking, and master life-saving actions.

3 of 4 sections completed

Learning Objectives

By the end of this module, students and school staff will be able to:

Explain why tectonic plates rupture and generate seismic wave vibrations.

Identify high-risk hazards in classrooms, corridors, and campus labs.

Execute the gold-standard "Drop, Cover, and Hold On" procedure in seconds.

Safely perform aftershock checks, injury assessments, and evacuation.

Lesson Outline

Learning Objectives

1. What is an Earthquake?

2. Before the Quake

3. Drop, Cover, Hold On

4. Aftershocks & First Aid

Done

Done

Done

In Progress

Up Next

1. What is an Earthquake?

An earthquake is an intense shaking of Earth's surface caused by a sudden slip on a geological

Ready to Test Your Knowledge?

QUIZ



Ready to Test Your Knowledge?

Score 70% or higher to earn +50 XP and unlock the "Earthquake Ready" badge!



Start Earthquake Quiz

← Back to Lesson • [Assessment Mode](#)

Earthquake Preparedness Certification Quiz



Earthquake Safety Standard 101

5 Multi-choice questions • Passing score: 70%

0 / 5 Answered

Question 1 of 5

+20 Points

1. What should you do immediately during an earthquake if you are indoors?

- ☐ A Run outside immediately through stairwells while shaking is active.
- ☐ B Stand under an open doorway or near exterior windows.
- ☐ C Drop to hands and knees, take Cover under a sturdy desk, and Hold On firmly.
- ☐ D Take the building elevator straight to the ground floor.



Good Effort! Keep Practicing

You scored 60%. Review the safety tips below and re-attempt to earn your certification.

3/5

CORRECT ANSWERS

2/5

INCORRECT

+60 XP

POINTS EARNED

Question 5 of 5

+20 Points

5. What is the full procedural sequence behind "Drop, Cover, and Hold On"?

- A Drop books on the floor, cover your mouth with a cloth, and hold open the door.
- B Drop to hands and knees, take Cover under a desk protecting head/neck, and Hold On to the shelter.
- C Drop down on your stomach in an open athletic field and hold hands.
- D Cover eyes with your palms and remain standing motionless in the corridor.

Exit Assessment

Submit Final Answers & View Score

Question-by-Question Review

1. What should you do during an earthquake?

✓ Correct (+20 pts)

Your Answer: Option C

Correct Answer: C. Drop, Cover, and Hold On

 **Explanation:** Drop, Cover, and Hold On protects the vital head and cervical spine while preventing falls caused by seismic acceleration.

2. Which location is safest during shaking?

✗ Incorrect

Your Answer: Option A

Correct Answer: B. Under a sturdy desk or interior table

 **Explanation:** Sturdy furniture shields occupants from collapsing ceilings, light ballasts, and ductwork.

3. What should you do after an earthquake?

✗ Incorrect

Your Answer: Option D

Correct Answer: B. Check for injuries and follow orderly evacuation to assembly area

 **Explanation:** Never ignite matches after earthquakes due to potential severed gas pipes. Evacuate via stairs to the muster zone.

MOCK SCENARIO

**DisasterEd**
Learn. Prepare. Stay Safe.

LEARNING PORTAL

Dashboard

Learning Modules

Quizzes

Mock Scenarios

MY ACHIEVEMENTS

My Progress

Badges

Certificates

CAMPUS & SAFETY

Resources

ACCOUNT

John Doe
Student

Search safety lessons, drills, protocols...

Student View

Teacher / Admin

John Doe
Central Valley College

ID

John Doe
Central Valley College

Emergency Response Simulator

Test your real-time decision-making skills in realistic high-stakes campus emergency simulations.

SIMULATION STATUS

0 of 4 Drills Completed

Scenario 1 Earthquake in Classroom

2nd Floor Science Building

Situation:

You are seated in your lecture hall when violent shaking begins without warning. Overhead light fixtures swing wildly, books slide off desks, and ceiling plaster begins cracking.

What is your immediate, safest course of action?

A

Run outside into the courtyard immediately while shaking occurs.

B

Stand directly next to exterior window glass to check outside.

C

Drop, Cover, and Hold On under your sturdy desk until shaking stops.

D

Sprint into the elevator to quickly reach the ground floor.

Scenario 1 Earthquake in Classroom

2nd Floor Science Building

Situation:

You are seated in your lecture hall when violent shaking begins without warning. Overhead light fixtures swing wildly, books slide off desks, and ceiling plaster begins cracking.

What is your immediate, safest course of action?

A

Run outside into the courtyard immediately while shaking occurs.

B

Stand directly next to exterior window glass to check outside.

C

Drop, Cover, and Hold On under your sturdy desk until shaking stops.

D

Sprint into the elevator to quickly reach the ground floor.

Safety Warning & Lesson:

Correct! "Drop, Cover, and Hold On" is the single most proven life-saving action during an earthquake. Running outside risks falling glass and masonry bricks. Elevators can lose power or jam.

Try Scenario Again

Next Scenario

MY PROGRESS

DisasterEd

Learn. Prepare. Stay Safe.

LEARNING PORTAL

Dashboard

Learning Modules

Quizzes

Mock Scenarios

MY ACHIEVEMENTS

My Progress

Badges

Certificates

CAMPUS & SAFETY

Resources

ACCOUNT

Search safety lessons, drills, protocols...

Student View

Teacher / Admin

JD

John Doe

Central Valley College

My Learning Progress

Track your preparedness curriculum completion, weekly activity hours, and quiz score analytics.

62%

OVERALL COMPLETION

4 / 8

MODULES COMPLETED

85%

AVG. QUIZ SCORE

8.5 hrs

TOTAL LEARNING TIME

Weekly Learning Activity

Study minutes spent across campus safety drills

Last 7 Days

100m

90m

80m

70m

60m

50m

40m

30m

20m

10m

0m

Mon

Tue

Wed

Thu

Fri

Sat

Sun

45m

60m

30m

80m

50m

95m

70m

Disaster Domain Mastery

Distribution of completed curricula

Geological

Fire Safety

Flood/Water

First Aid

Geological (85%)

Fire (60%)

Flood (40%)

View Certificates

Badges Showcase

CERTIFICATE

My Certifications History

Course Title	Certificate No.	Issued Date	Score	Status	Actions
Comprehensive Disaster Preparedness & Earthquake Safety Standard FEMA-Aligned Curriculum	DED-2026-8941	Sep 01, 2026	85% (Distinction)	Verified	Print

My Safety Certification

Verified institutional certificates of accomplishment in disaster response and campus resilience.

Print / Download Certificate PDF



My Certifications History

Course Title	Certificate No.	Issued Date	Score	Status	Actions
Comprehensive Disaster Preparedness & Earthquake Safety Standard FEMA-Aligned Curriculum	DED-2026-8941	Sep 01, 2026	85% (Distinction)	Verified	Print

BADGES

Search safety lessons, drills, protocols...

Student View

Teacher / Admin

John Doe
Central Valley College

Preparedness Badges & Achievements

Earn official digital micro-credentials by excelling in emergency simulations and assessment tests.

4 of 6 Badges Unlocked

Earthquake Ready
Demonstrated mastery of the Drop, Cover & Hold On procedure and structural damage assessment.
Earned on Aug 28, 2026

Fire Safety Hero
Completed laboratory fire safety simulation and smoke crawl evacuation with 100% precision.
Earned on Aug 30, 2026

Flood Awareness
Identified flash flood warning indicators and mapped safe high-ground escape routes.
Earned on Sep 01, 2026

First Aid Beginner
Learned how to apply emergency pressure bandages, CPR rhythms, and recovery positioning.
Earned on Sep 02, 2026

Quiz Master
Achieve top marks (85%+) on 3 consecutive emergency safety module assessments.
Criteria Progress 2 / 3 Quizzes

Disaster Prepared
Complete all core beginner curriculum modules and participate in the live interactive campus drill.
Overall Progress 75%

RESOURCES

DisasterEd

Learn. Prepare. Stay Safe.

LEARNING PORTAL

Dashboard

Learning Modules

Quizzes

Mock Scenarios

MY ACHIEVEMENTS

My Progress

Badges

Certificates

CAMPUS & SAFETY

Resources

ACCOUNT

John Doe

Student

Search safety lessons, drills, protocols...

Student ViewTeacher / Admin

JDJohn Doe

Central Valley College

Campus Emergency Resources & Toolkits

Quick reference guides, interactive survival kit checklists, and direct national emergency dispatch hotlines.

Print Guide

Emergency Hotlines & Campus Dispatch

24/7 Priority Lines

CAMPUS POLICE & SECURITY

911 / (555) 019-9911

NATIONAL DISASTER AGENCY

1-800-621-3362

CAMPUS HEALTH & MEDICAL

(555) 019-4325

POISON CONTROL CENTER

1-800-222-1222

Interactive Go-Bag / Emergency Kit Checklist

Check off items as you prepare your classroom or home survival backpack.

7/10 Items Packed

☒ Bottled Drinking Water (3-Day Supply)

☒ Non-perishable Energy Food & Bars

☒ First Aid Kit & Antiseptic Bandages

☒ LED Flashlight & Extra AA/AAA Batteries

Campus Evacuation Guidelines

1. Remain Calm & Listen:

Pay immediate attention to public address (PA) announcements and faculty marshals.

2. Never Use Elevators:

Power faults can cause entrapment. Always use marked fire exits and perimeter stairs.

3. Stay Low Under Smoke:

Toxic thermal gases rise to ceilings. Crawl on hands and knees where air is cleaner.

4. Proceed to Assembly Area:

Report to the football stadium lawn and check in with your class roll representative.

TEACHER/ADMIN

DisasterEd

Learn. Prepare. Stay Safe.

LEARNING PORTAL

Dashboard

Learning Modules

Quizzes

Mock Scenarios

MY ACHIEVEMENTS

My Progress

Badges

Certificates

CAMPUS & SAFETY

Resources

ADMINISTRATION

Dr. Sarah Jenkins

Teacher

Search safety lessons, drills, protocols...

Student ViewTeacher / Admin

SJDr. Sarah Jenkins

Central Valley College

Switched view to Dr. Sarah Jenkins (Teacher)

Faculty & Campus Safety Administration

Campus Preparedness Management Portal

Add ModuleCreate QuizManage StudentsView Reports

250

TOTAL STUDENTS ENROLLED

187

ACTIVE THIS WEEK

78%

AVERAGE QUIZ SCORE

1,240

MODULES COMPLETED

Student Completion Trends

Monthly module completions across campus departments

Academic Year 2026

Quiz Score Distribution

Cohort grade performance breakdown

 **DisasterEd**
Learn. Prepare. Stay Safe.

LEARNING PORTAL

Dashboard

Learning Modules

Quizzes

Mock Scenarios

MY ACHIEVEMENTS

My Progress

Badges

Certificates

CAMPUS & SAFETY

Resources

ADMINISTRATION

SJ Dr. Sarah Jenkins
Teacher

Search safety lessons, drills, protocols...

Student View Teacher / Admin

SJ Dr. Sarah Jenkins
Central Valley College

Disaster Learning Modules

Interactive, scientifically verified curricula designed for primary, secondary, and higher education campus safety.

8 Modules Available

All CategoriesGeological HazardsWeather & StormsFire & StructuralFirst Aid & Health

Curriculum Standard: FEMA / Red Cross Aligned

 Beginner

Earthquake
Master seismic tremors, structural vulnerability assessment, and the vital "Drop, Cover, and Hold On" drill.
20 Mins **75% Completed**
Continue Learning

 Intermediate

Flood
Understand flood warning signals, dangerous flash surge currents, safe high ground, and contamination hazards.
25 Mins **40% Completed**
Continue Learning

 Beginner

Fire
Learn fire hazard mitigation, the PASS fire extinguisher technique, crawling under smoke, and assembly routines.
15 Mins **60% Completed**
Continue Learning

 Intermediate

Cyclone
Analyze meteorological track maps, severe barometric pressure drops, and reinforce interior shelter spaces.
20 Mins **0% Completed**
Start Learning

 Advanced

 Advanced

 Beginner

 Beginner